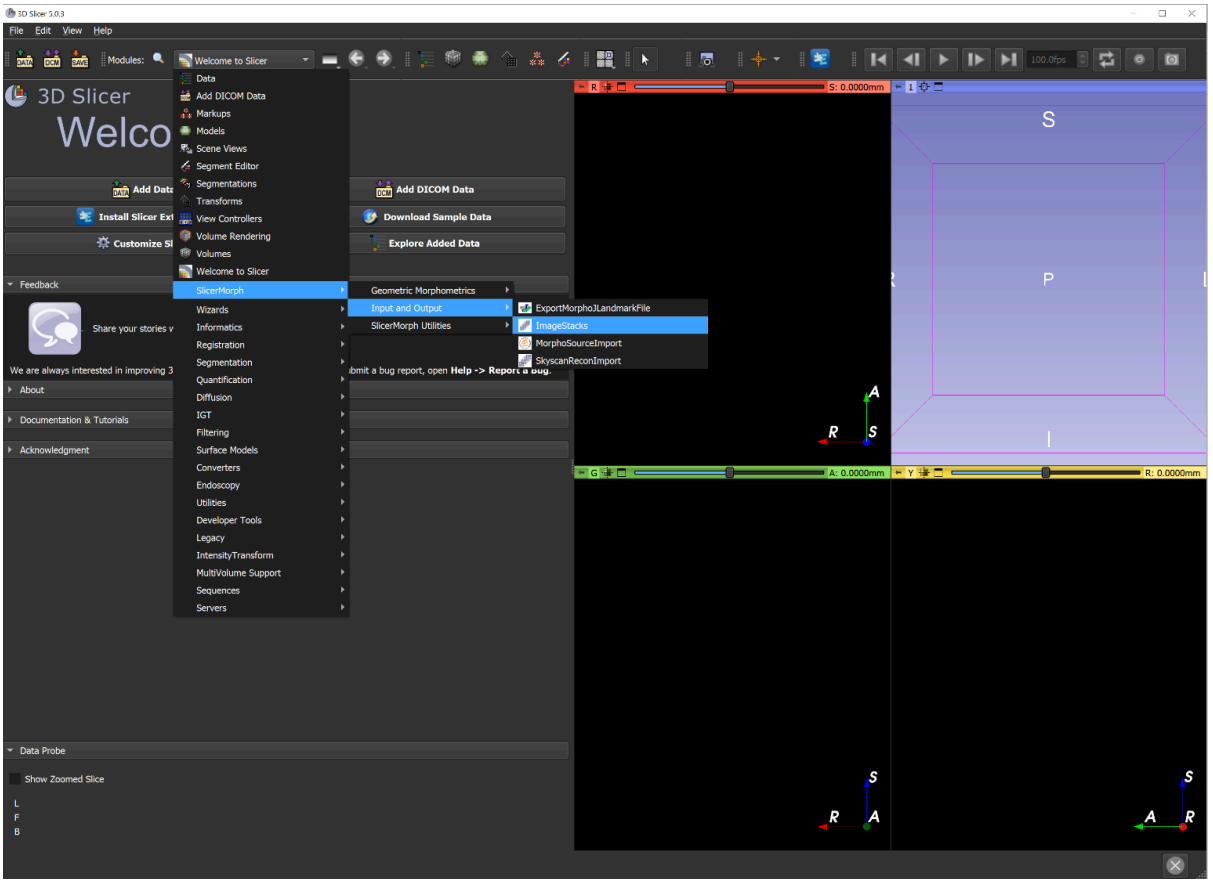
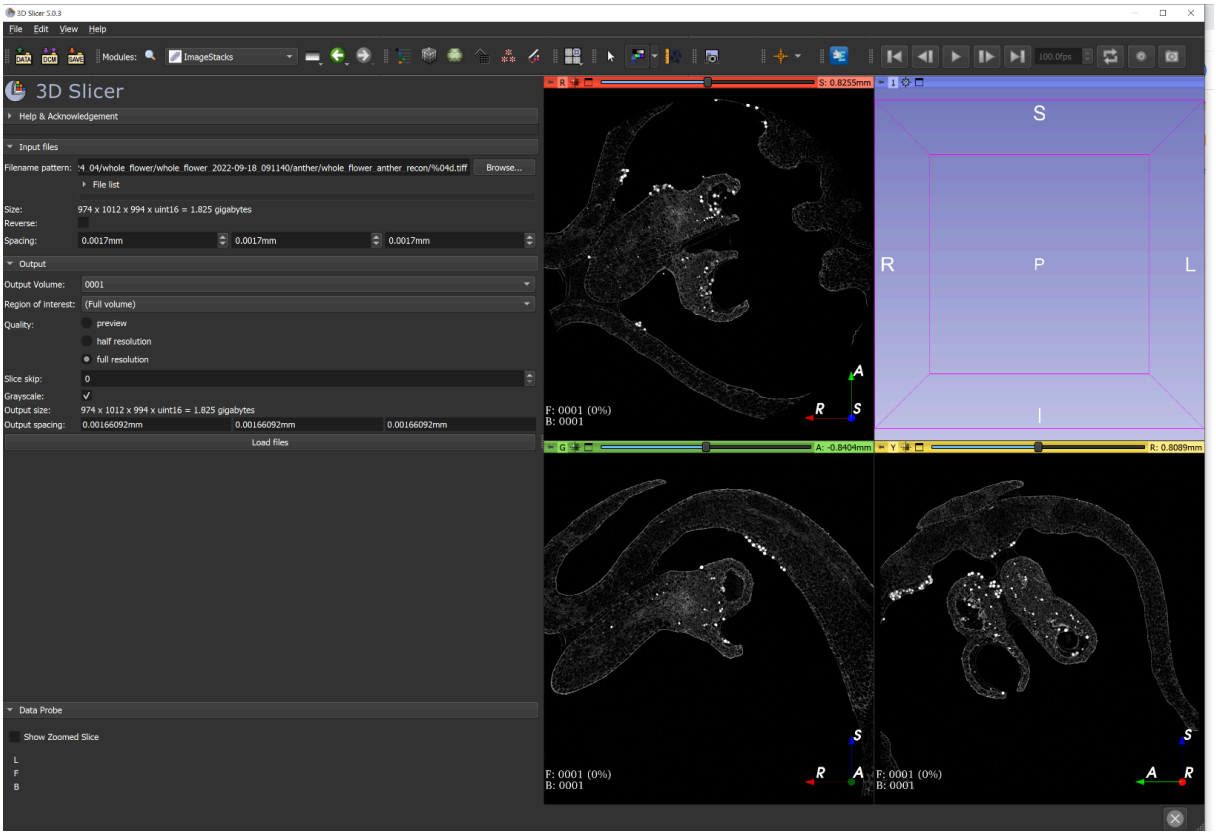
Data import into 3D slicer

Last edited 2 December 2022 by K. Wolcott

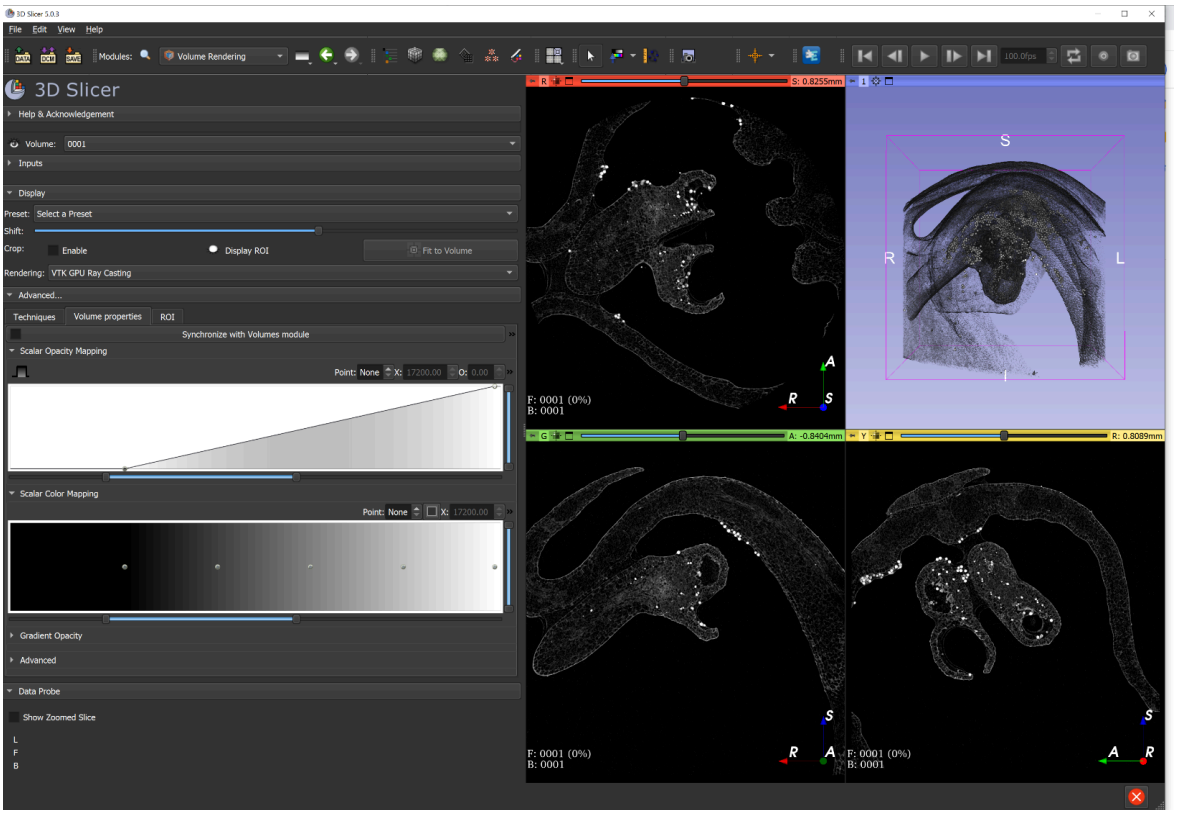
1. Open 3D Slicer
2. Import CT data
    a. Modules
        i. Slicermorph module
            1. Input and Output
                a. Image stacks



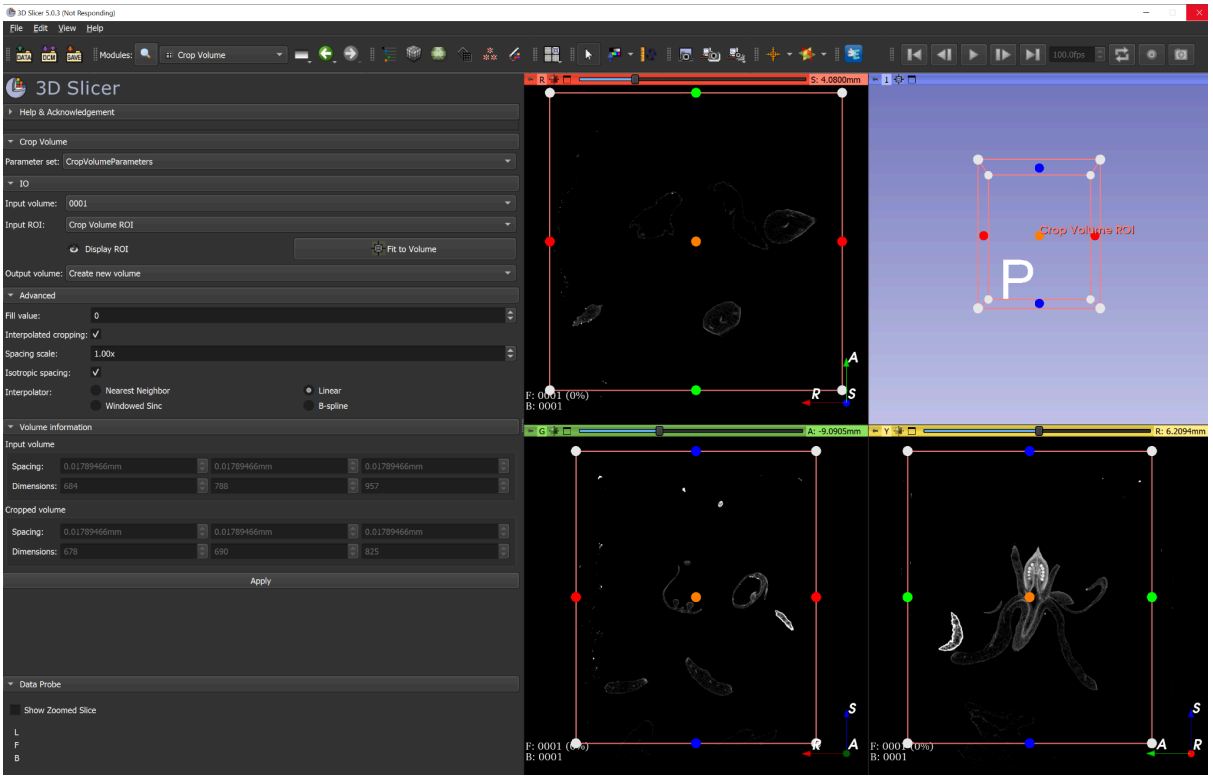
b. Open reconstructed image stack at half resolution and skip up to 1-2 slices for a final dataset size of <2GB to reduce processing speeds



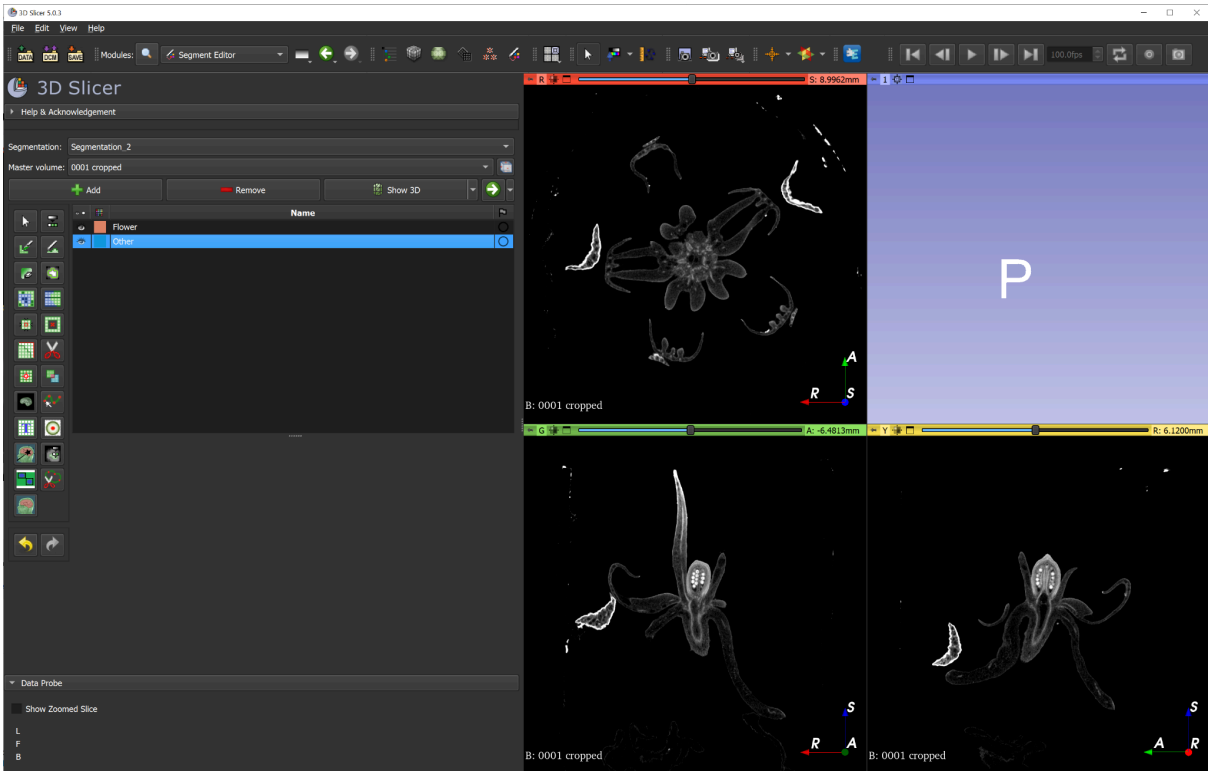
3. Visualize dataset as 3D volume
    i. Go to Volume Rendering module and adjust “Shift” until you can see the volume



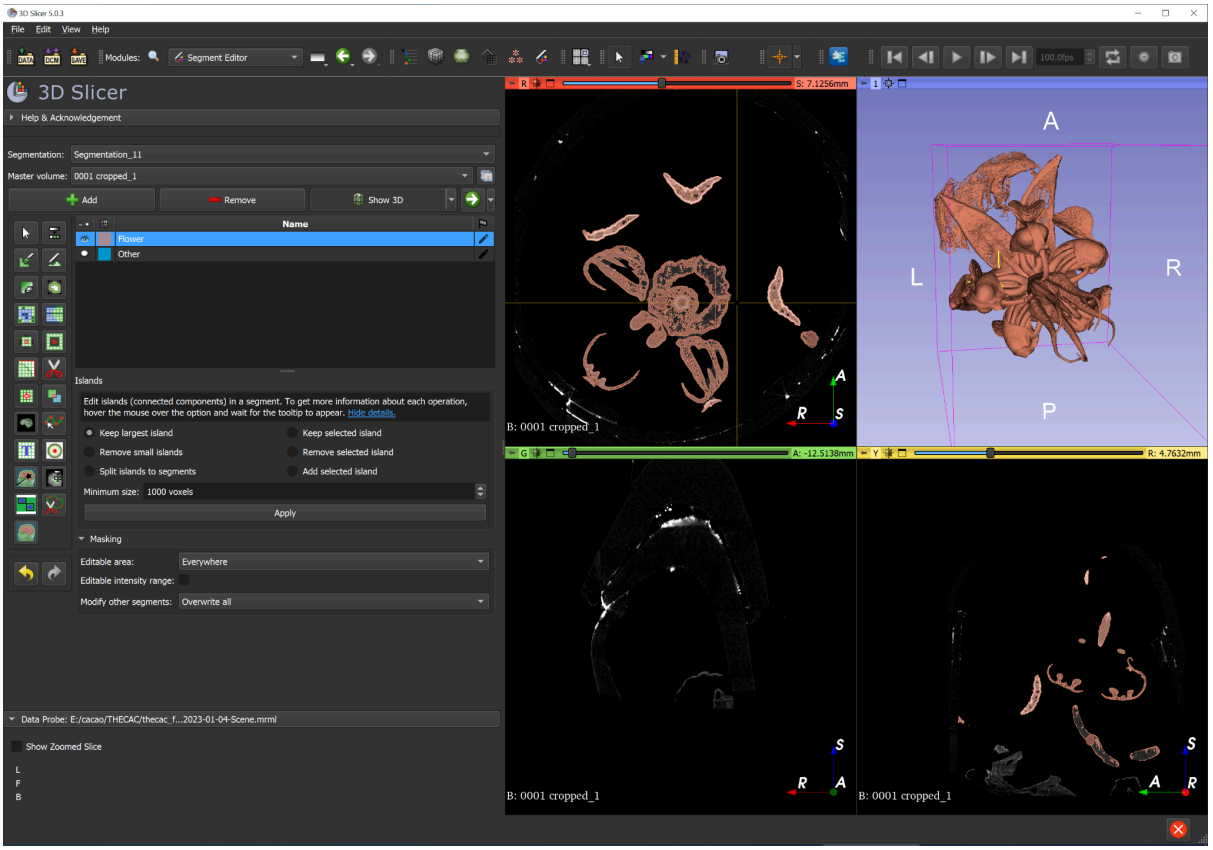
4. Crop Volume to reduce file size
    a. Go to Crop Volume Module
    b. Go through XYZ planes and adjust ROI to include entire flower, but minimize blank space around edges
    c. Click “Apply”
    d. Click eye icon by Input ROI to hide ROI
    e. See tutorial https://www.youtube.com/watch?v=BJolexIvtGo



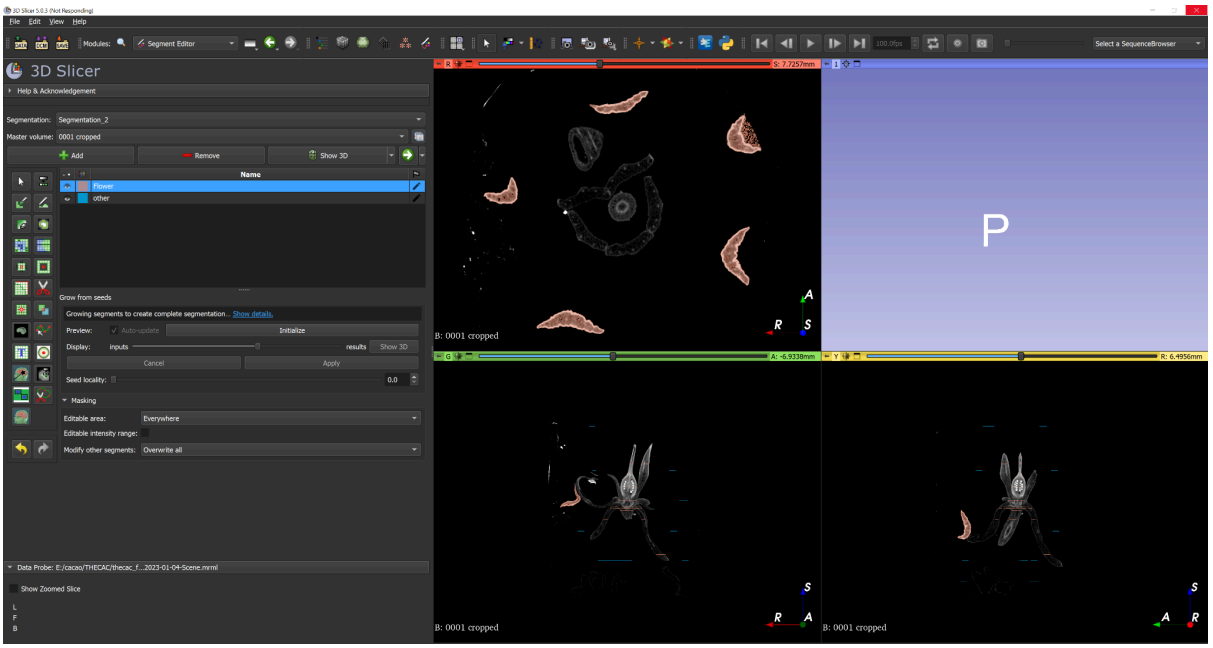
5. Segment volume
    a. Click segment editor icon on top toolbar
    b. Add segments for “Flower” and “Other”
        i. Color segments
            1. Flower - Organ
            2. Other - Vein



c. Use thresholding to maximize flower coverage
d. Use islands -> keep largest island to get rid of noise around specimen holder edges

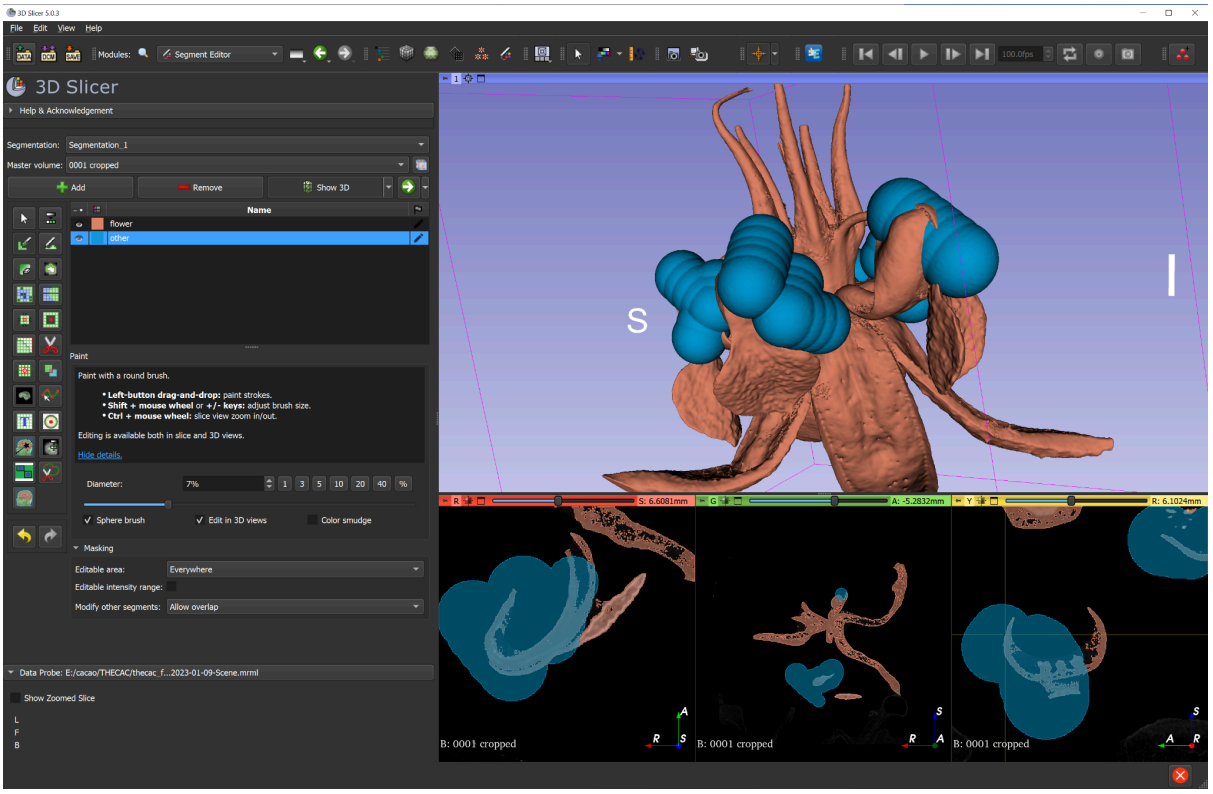


e. Use paintbrush and grow from seeds to paint sections of other and flower throughout volume; then make fine scale adjustments using level tracing and smoothing

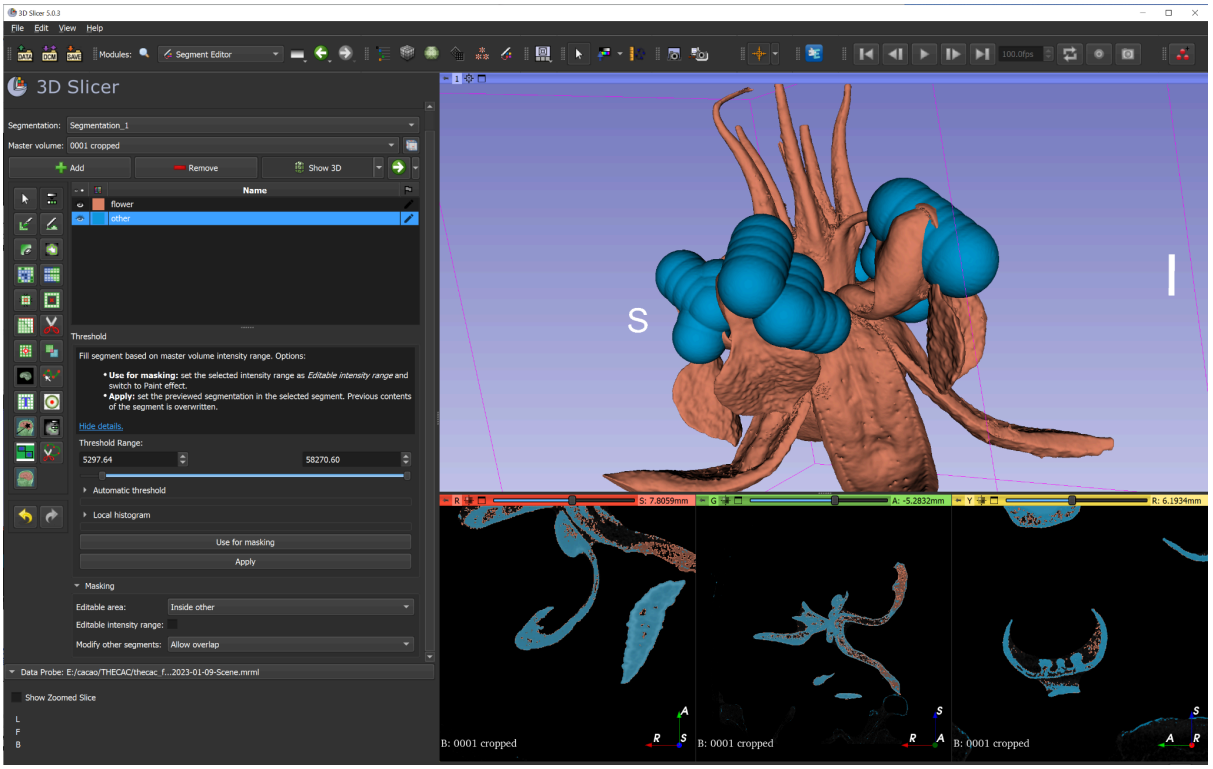


f. If petals or sepals have lots of holes from differential contrast when compared to other parts, use masking and thresholding

i. Paint in 3D to “other” (editable area: everywhere; modify other segs: allow overlap)



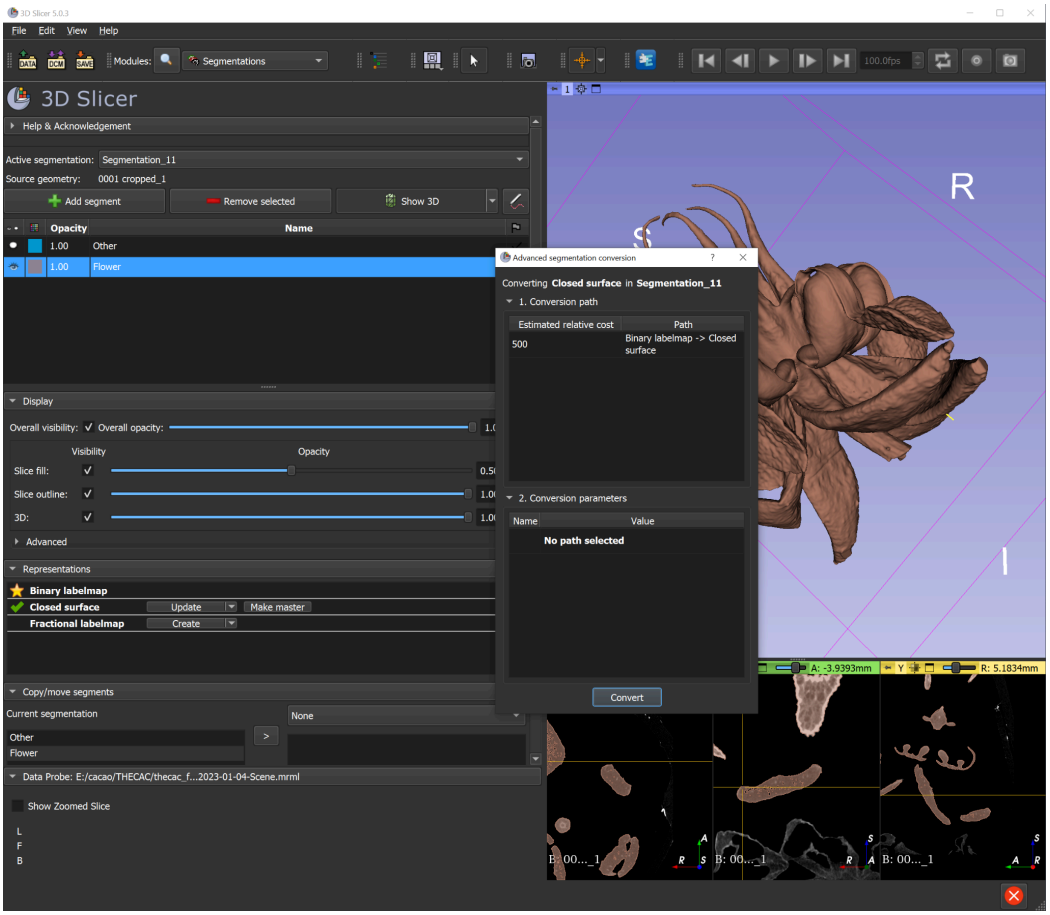
ii. Use thresholding within “Other” then click “flower” and “apply”



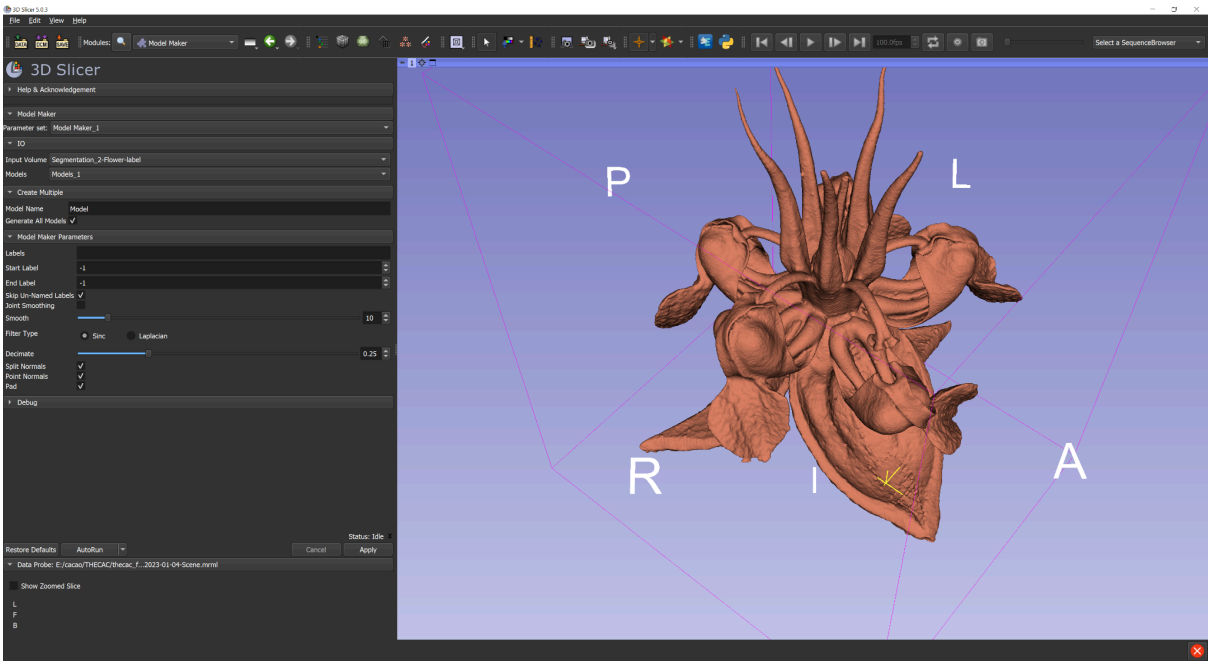
iii. Hide “other” and inspect changes

g. Finish by removing islands again (otherwise little specs might get auto landmarked later)
h. See tutorial https://www.youtube.com/watch?v=tJMGe3FMTk0
i. https://github.com/SlicerMorph/Tutorials

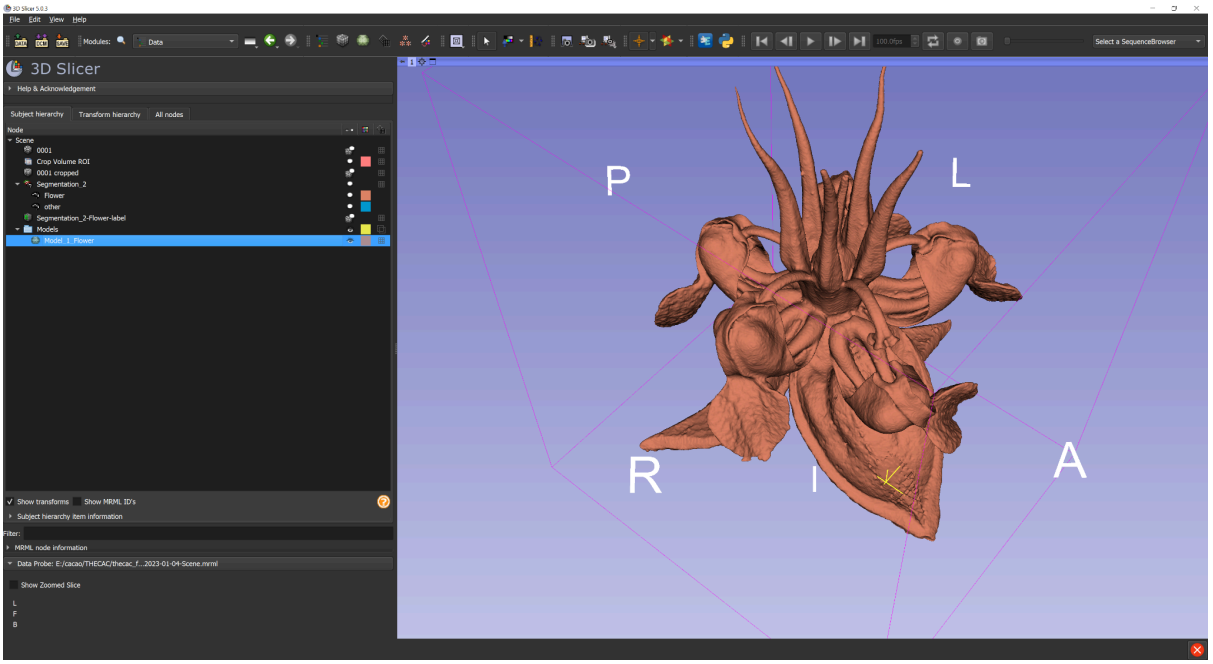
6. Save segmentations
   a. Go to Segmentations
   b. Delete “Other”
   c. Representations -> Closed surface
      i. Click on conversion path contents (“500 Binary Labelmap”)
      ii. Convert once you see parameters listed in 2
   d. Export/Import models and labelmaps
      i. Export labelmap
      ii. Export models



7. Generate surface
    a. Open Model Maker Module
    b. Create model



    c. Export model as .ply
        i. Go to Data Module
        ii. Right click on model file
            1. Export as .ply


3D Slicer
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